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MARCH 2025 **ADVANCED LEVEL**

Subject title	Physics
Subject code	0780
Paper number	Paper 2

Duration: Three Hours

Answer ALL questions

Section I is designed to be answered in 1hour, Section II in 30 minute and Section III in 1hour.

You are advised to divide your time accordingly

You are reminded of the necessary for good English and orderly presentation in your answers.

In calculation you must show all the steps in your writing giving your answer at each stage.

Calculators and formulae booklets are allowed

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SECTION I

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Answer all questions
(one hour)

1. Figure 1 shows a circuit made up of two capacitors which can be charged by the battery of 8.0V and discharged through the series resistors.

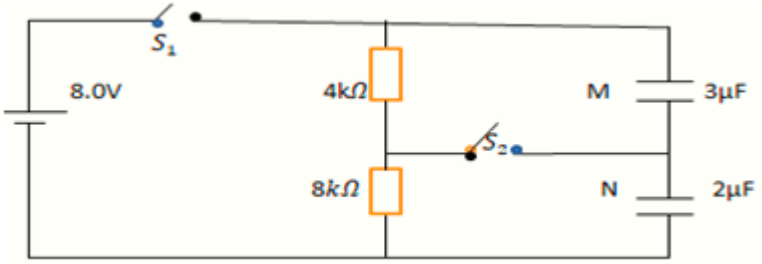


Figure 1

The time constant (τ) of the circuit can be defined by

$$\tau = RC.$$

Where R represents the total resistance of the circuit and C is the equivalent capacitance of the circuit.

a) Explain the fact that homogeneity of physical equations is not sufficient for the correctness of the equation.
b) Show that the above equation is homogeneous.
c) Determine the time constant for the above circuit if S_2 stays open throughout charging and discharging process of the capacitors and the charge stored in the capacitors. (7 marks)

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2. The circuit bellow shows resistor and cells connected in a circuit.

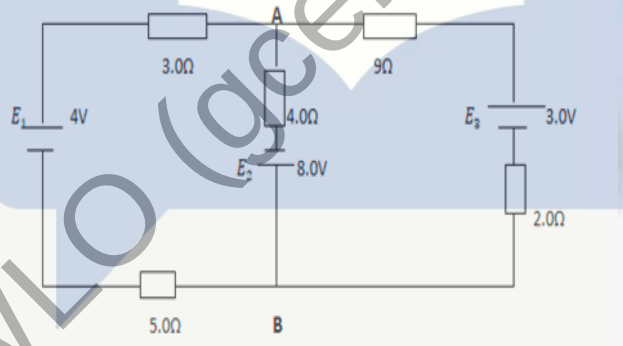


Figure 2

Calculate the
a) The currents through each of the resistors are 3.0Ω, 4.0Ω and 9.0Ω.
b) The potential difference between A and B.
c) Distinguish between potential difference and electromotive force. (6marks)

---

3. (a) define gravitational potential at a point.

(b) the gravitational potential Ø at a distance r from point mass M is given by the expression.

$$\emptyset = -\frac{GM}{r}$$

Where G is the gravitational constant.
i) Explain the significance of the negative sign in this expression.

ii) A spherical planet may be assumed to be and isolated point mass with it mass concentrated at the center. A small mass m is moving near to, a normal to, the surface of the planet. The mass moves away from the planet through a short distance h, state and explain why the change in the gravitational potential energy ΔE of the

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mass is given by the expression $\Delta E = mgh$ where g is the acceleration of free fall. **(6marks)**

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3. A point source of gamma radiation has a half – life of 45minutes. The initial count rate recorded by a GM tube placed 3.0 m from the source is $864s^{-1}$. The distance between the counter and the source is altered. After 1.5 hours the count rate recorded is 6 s^{-1}

a) Explain the meaning of the underlined phrase.

b) Calculate the new distance between the counter and the source. (5marks)

---

4. (a) Distinguish clearly between diffraction and interference.

(b) Monochromatic light of wave length $5.0 \times 10^{-7}m$ falls normally on a diffracting grating which has 425 lines per milometer.

i) What is the highest order of diffraction?

ii) Fine the angular separation between the first and the third ordered lines **(6marks)**

Answer EITHER 6(a),(b) and (c) OR 6(d), (e) and (f).

EITHER 6(a),(b) and (c)

6. (a) (i) Distinguish between thermionic emission and photoelectric effect. **(2marks)**

(ii) describe an experiment to measure the value of the plank constant. Your description should include a diagram, procedure, measurement, precautions and how you will use the measurement to arrive at a conclusion. **(8marks)**

(b) (i) use the equation of and ideal gas to show that the average translational kinetic energy of the gas molecules in the container is

$$E = \frac{3}{2}kT.$$

(ii) Explain what is meant by an ideal gas and give an example of an ideal gas **(6marks)**

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c) Two particles carrying charges $Q_1 = 4.0\mu C$ *and* $Q_2 = -3\mu C$ are place distance of $5.0 \times 10^{-2}m$ *apart*.

i. Sketch the electric field lines between the charges Q_1 and Q_2

ii. Calculate the point on the line passing through the two particles at which the resultant field is zero. **(4marks)**

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OR 6(d), (e) and (f)

6 (d) (i) Define specific latent heat of vaporisation;

(ii) Describe and experiment to determine the specific latent heat of fusion of ice; your description should include a diagram, procedure, observation and how you would use the collection data to determine the specific latent heat of fusion.

(iii) 0.03 kg of ice and 0.20kg of water at 0 °C are in a container .Steam at 100°C is passed in until all the ice melted and the temperature in the container rise to 50°C .How much water is now in the container.

(e) i) A future fusion reactor might use the reaction $^{2}_{1}H + ^{2}_{1}H \rightarrow ^{4}_{2}He + energy$ to produce useful energy from the following data. Calculate the number of reactions required to produce 1J of energy.

ii) Calculate the mass of $^{2}_{1}H$ required to provide 1J of energy.
(mass of $^{2}_{1}H$ =2.0136U, 1U=1.661x10^{-27}kg, $^{4}_{2}He = 4.0015U$)

(f) i) The energy levels of hydrogen atom are given by the equation

$$E_n = \frac{-2.16\,x10^{-18}}{n^2}\,J$$

Determine the ionisation energy of the atom and the wavelength of the H_α line, which arises from transition between n=3 and n=2 levels.

ii) Elections may be emitted from a surface by thermionic emission or by photoelectric emission. Distinguish between the two.

SECTION II (30minutes)

DATA ANALYSIS

7. The resistance R of a thermistor is related to its absolute temperature, T by the equation

$$R = R_0 \exp\left(\beta\left[\frac{1}{T} - \frac{1}{T_0}\right]\right) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots 1$$

Where $R_0 = 1.0\,x\,10^6\Omega$ is the resistance at temperature T_0 and β is a constant under certain conditions.

In an experiment the value of R and θ obtained are displaced in table 1 below

R/MΩ	2.0	3.0	5.0	7.0	11.0	19.0	30.0	46.0	64.0	95.0
Θ °C	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0

a) Plot a suitable graph to determine the values of β and T_0 **(13marks)**

b) Use your graph to determine the values of β and T_0 **(5marks)**

c) Determine the unit of $\frac{\beta}{T_0}$ **(2marks)**

(Total = 20marks)

SECTION III (1hour)

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OPTIONS

Answer any two questions from the four options

OPTION 1: ENERGY RESOURCES AND ENVIRONMENTAL PHYSICS

8. a) i) Distinguish between conventional energy source and non-conventional energy source stating an example in each case.

(3marks)

ii) Energy is conserved explain why there is still energy crisis despite this conservation of energy**.**

(1mark)

b) At a tidal power station, water is trapped over an area of 250 km^2 when the tide is 5.0 m above the power station turbines . The trapped water is released gradually over a period of 6hours. Calculate

i) the mass of the trapped water

ii) the average loss of potential energy per second of this trapped water when it is released over a period of 6hours. (Density of sea water= 1050 kg m^{-3})

iii) Calculate the power output of the power station if the combine efficiency of the turbines and generator is 75%.

(6marks)

c) i) Define climate change

ii) Name two greenhouse gases and explain an environmental effect of any of these gases.

iii) State one impact of whether on aviation and space exploration. **(5marks)**

(Total =15marks)

OPTION 2: COMMMUNICATION

9. (a) Figure six shows a circuit diagram of a simple radio receiver.

Which of the components in the above circuit:

(i) Remove the radio frequency (R.F) carrier to leave the audio signal?

(ii) Select the required R.F signal?

(iii) Changes the R.F to give a non-zero average signal?

(iv) Carries many R.F signals at all times

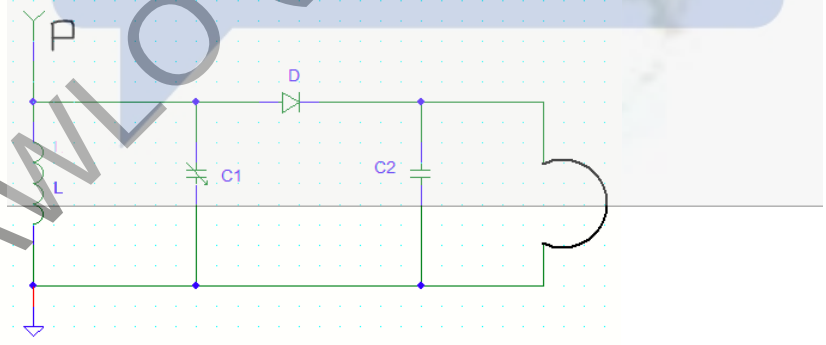


Figure 6

(b) Figure 7 shows the diagram of superheterodyne radio receiver.

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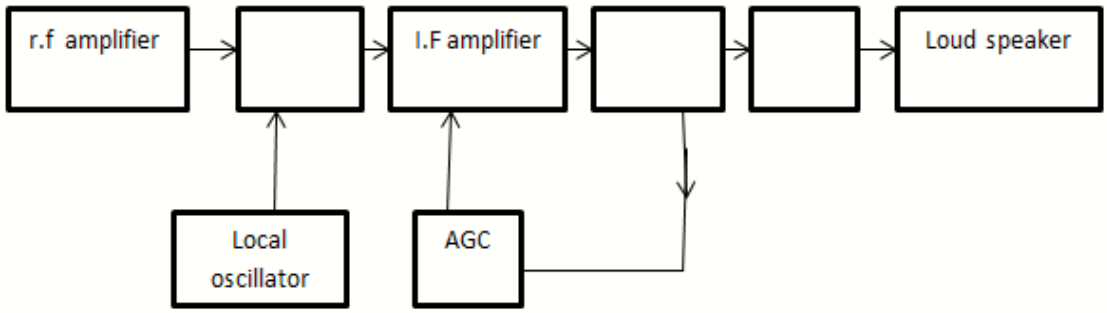


Figure 7

(i) What is mean by superheterodyning?
(ii) Copy and complete the diagram by adding the names of the unlabelled blocks.
(iii) Explain the purpose of the automatic game control (AGC) subsystem.
(iv) The radio receiver receives a signal on a frequency of 1215 kHz. If the local oscillator frequency is 1670 kHz, calculate the value intermediate frequency (IF). **(7marks)**
(c) (i) What is a base station?
(ii) Briefly explain the role of a base station during the making of a call from a mobile phone handset. **(4marks)**

OPTION 3: ELECTRONICS

10. (a) (i) what is doping?

(ii) Distinguish between N-type and P-type doping **(3marks)**
(b) Figure 8 shows a simple circuit with a device, D connected in series with the battery and a resistor

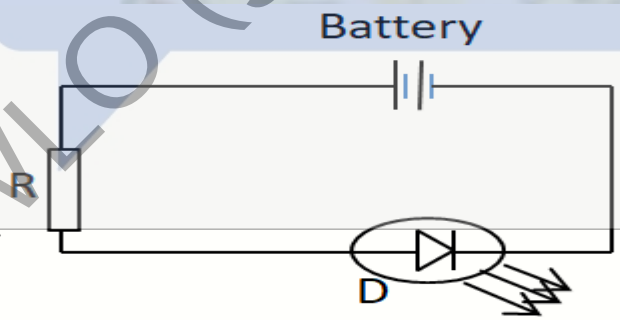


Figure 8

(i) What is the name of the device D?
(ii) What is the function of R?
(iii) Briefly explain how the current in the circuit can be increased
(iv) What value of series resistor is required to limit the current through D to 25 mA with a forward voltage drop of 1.8V when connected to a 9.0V supply? **(6marks)**

(c) Figure 9 shows a transistor circuit with a current gain of 150 and V_{BE} of 0.7V

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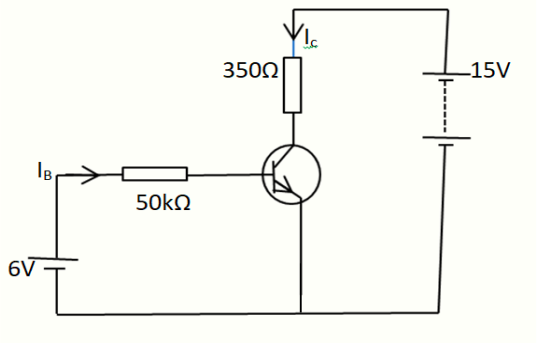


Figure 9

(i) Determine the base and collector currents I_B and I_C
(ii) Determine the quiescent (operating) point.
(iii) Sketch the DC load line **(6marks)**

OPTION 4: MEDICAL PHYSICS

11. (a) A girl with defective eyes has a far point of 280 cm.
(i) State the type of the eye defect she is suffering from.
(ii) Determine the type of lens needed to correct this defect.
(iii) Draw a ray diagram to show how this defect can be corrected. **(5marks)**

(b) (i) Briefly explain how a CT scanner works.
(ii) Give one advantage of using a CT scan compared to MR or ultrasound.
(iii) X-rays with initial intensity 30.0Wm^{-2} are used to image bone with density of 1900 $kg\ m^{-3}$ and mass attenuation coefficient of 0.133 m^2kg^{-1}. Find the linear attenuation coefficient of the bone and the intensity of the X-rays after they pass through 0.690 cm of bone. **(7marks)**

(c) In a gastroscopy, an endoscope is sent down the patient's throat to see inside their digestive tract. State the role that the following will play in a gastroscopy:
(i) Non-coherent optical fibre bundle.
(ii) Coherent optical fibre bundle.
(iii) State what an ECG trace shows. **(3marks)**

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